

CRNI VRH, Serbia

WMO: 132890

Lat: 44.1297N

Lon: 21.9642E

Elev: 3409

StdP: 12.97

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	4.1	9.2	0.4	6.4	5.9	5.7	8.3	11.9	32.3	2.9	28.3	17.1	18.1	300	0.613

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	13.0	81.1	63.4	78.1	62.5	75.4	62.1	66.5	75.4	64.8	73.9	63.5	72.4	6.7	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
63.4	99.2	70.9	61.7	93.4	69.3	60.1	88.2	67.5	33.1	75.7	31.8	74.0	30.6	72.5	73.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
25.5	22.6	20.0	0.1	85.6	4.8	5.1	-3.3	89.2	-6.1	92.2	-8.8	95.0	-12.3	98.7		
			-0.6	69.3	4.7	1.9	-4.0	70.6	-6.7	71.7	-9.3	72.8	-12.7	74.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	45.6	26.3	28.3	35.1	45.1	53.8	60.1	64.2	64.8	55.3	46.1	37.5	29.0
	DBStd	16.10	10.25	10.45	9.60	8.57	7.38	7.17	6.74	6.93	7.79	9.34	9.36	9.45
	HDD50	3317	735	607	467	190	46	7	1	1	35	190	387	651
	HDD65	7298	1200	1027	926	599	351	177	98	90	303	587	825	1115
	CDD50	1702	0	1	6	42	165	311	441	459	195	69	12	1
	CDD65	206	0	0	0	1	5	31	73	82	13	1	0	0
	CDH74	896	0	0	0	1	14	93	359	393	36	0	0	0
	CDH80	147	0	0	0	0	0	6	67	72	2	0	0	0
Wind	WSAvg	10.6	12.6	12.4	11.5	10.6	9.6	8.8	8.8	8.8	9.5	10.6	11.7	12.6
Precipitation	PrecAvg	29.6	1.9	1.8	1.9	2.7	3.1	3.3	2.6	2.1	2.8	2.8	2.3	2.3
	PrecMax	42.2	3.9	3.7	3.6	5.6	6.3	8.5	6.7	6.4	6.1	5.7	5.3	6.1
	PrecMin	19.5	0.5	0.3	0.5	0.5	1.1	0.7	0.4	0.1	0.4	0.6	0.3	0.2
	PrecStd	5.7	1.0	0.9	0.9	1.1	1.5	1.8	1.7	1.5	1.7	1.5	1.2	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.2	55.9	61.1	71.0	75.7	80.5	86.7	86.8	78.4	69.8	62.9	54.7
		MCWB	41.2	44.6	47.2	53.8	59.2	65.0	62.0	63.5	61.7	57.0	49.7	43.2
	2%	DB	47.0	50.2	56.2	65.2	70.5	76.7	81.1	81.6	73.3	65.6	57.2	48.6
		MCWB	39.2	41.0	44.4	51.4	57.3	64.1	63.2	63.4	58.9	55.0	47.3	41.4
	5%	DB	43.7	45.9	52.4	60.9	67.5	73.7	77.8	78.5	69.8	62.1	53.5	45.2
		MCWB	38.3	39.2	42.5	49.2	56.4	62.8	62.8	62.3	58.1	53.3	46.2	39.8
	10%	DB	40.5	42.3	48.6	57.0	64.7	71.2	74.8	75.4	66.4	58.8	50.3	42.2
		MCWB	36.6	37.3	41.1	47.2	55.3	61.4	62.2	61.6	56.6	51.5	44.9	38.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.0	45.5	48.3	55.6	61.7	68.4	67.5	69.0	63.4	58.9	51.7	46.1
		MCDB	49.0	53.9	58.6	66.3	71.3	75.7	76.9	78.6	74.2	68.1	58.6	50.5
	2%	WB	40.7	42.3	45.6	52.6	59.4	66.1	65.9	66.3	60.7	56.2	49.4	42.4
		MCDB	45.2	48.3	53.7	62.4	67.4	73.3	75.3	76.2	70.9	63.0	54.7	47.3
	5%	WB	38.8	39.9	43.5	50.4	57.7	64.1	64.6	64.4	58.9	54.3	47.3	40.5
		MCDB	42.8	44.7	50.5	59.0	65.4	71.5	73.8	74.3	67.2	60.4	51.5	44.5
	10%	WB	36.8	37.6	41.5	48.3	55.9	62.2	63.2	62.8	57.3	52.4	45.4	38.3
		MCDB	40.2	42.0	47.7	55.4	63.0	69.6	72.1	72.4	64.5	57.8	49.2	41.9
Mean Daily Temperature Range	5% DB	MDBR	8.8	9.1	10.7	12.3	11.8	12.1	13.2	13.0	11.1	10.0	8.8	8.3
		MCDBR	12.8	14.2	15.7	17.0	14.5	14.3	16.3	15.6	14.7	14.4	13.6	12.4
		MCWBR	8.0	8.5	9.0	8.9	8.6	8.8	8.0	7.6	7.7	8.1	7.8	8.0
	5% WB	MCDBR	11.9	12.8	14.3	14.7	13.4	13.1	14.2	14.4	13.3	13.0	11.8	11.7
		MCWBR	8.3	8.4	9.0	8.4	8.9	8.9	8.1	8.3	7.9	8.0	7.6	8.4
Clear-Sky Solar Irradiance	taub	0.297	0.320	0.351	0.398	0.386	0.380	0.386	0.398	0.377	0.345	0.319	0.296	
	taud	2.385	2.315	2.322	2.202	2.278	2.321	2.300	2.268	2.311	2.409	2.440	2.420	
	Ebn at Noon	262	274	280	274	280	282	279	271	267	262	252	251	
	Edh at Noon	24	31	35	43	41	40	40	40	36	28	23	22	
All-Sky Solar Radiation	RadAvg	421	679	1056	1432	1735	1957	1983	1771	1248	832	499	362	
	RadStd	59	94	141	164	169	184	147	168	119	115	62	53	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (56 neighbors)	+1.01	N/A	N/A	N/A	+0.79	+1.17	N/A	N/A	N/A	N/A
		+1.17	N/A	N/A	N/A	+0.65	N/A	N/A	-301	+193	N/A

Nomenclature: See separate page